

Word Problems & Applications

Module 4 Test

Grade 9 • MTH1W

Time: 60 minutes • Total: 70 marks

Name: _____

Date: _____

Instructions: Show all work for full marks — a 'let' statement and an equation earn method marks. Marks for each question are shown in brackets. Calculators are permitted unless your teacher states otherwise.

Section A — Knowledge & Understanding

1. Write an expression for 'a number x increased by 9'. [1]

2. Write an expression for 'the product of 6 and a number n '. [1]

3. Write an equation for 'a number decreased by 5 is 12'. [1]

4. Find 50% of 84. [1]

5. A car goes 40 km in 1 hour. Find its speed. [1]

6. If the first of two consecutive integers is n , write the second. [1]

7. One side of a square is s . Write an expression for its perimeter. [1]

8. Write an expression for 'twice a number, plus 3'. [1]

9. A shirt costs \$c. Write an expression for the cost of 5 shirts. **[1]**

10. The sum of a number and 7 is 20. Find the number. **[2]**

11. Three times a number is 27. Find the number. **[2]**

12. Find 25% of \$48. **[2]**

13. A runner runs 30 km in 3 hours. Find the average speed. **[2]**

14. A rectangle is 6 cm long and 4 cm wide. Find its perimeter. **[2]**

15. A number is doubled and increased by 4 to give 16. Find the number. **[2]**

Section B — Application

16. Four times a number, decreased by 7, is 29. Find the number. **[2]**
17. The sum of two consecutive integers is 45. Find them. **[2]**

18. A taxi charges \$4 plus \$2 per km. A trip cost \$20. How many km was it? **[2]**
19. A 400 mL solution is 15% salt. How many millilitres of salt does it contain? **[2]**
20. A jacket priced \$90 is discounted 30%. Find the sale price. **[2]**
21. A boat travels 96 km at 24 km/h. How long does the trip take? **[2]**
22. The sum of three consecutive integers is 72. Find them. **[3]**
23. After 13% tax is added, a bill is \$79.10. Find the price before tax. **[3]**

24. Two angles are complementary (sum to 90°). One is 4 times the other. Find both angles. **[3]**

Section C — Thinking

25. A phone plan costs \$30 per month plus \$0.05 per minute. One month's bill was \$47.50. How many minutes were used? **[4]**

26. The sum of three consecutive odd integers is 63. Find the integers. **[4]**

27. Coffee worth \$7/kg is blended with coffee worth \$13/kg to make 15 kg of blend worth \$9/kg. How many kilograms of each type are used? (Very hard.) **[5]**
28. A rectangle's length is 3 m more than twice its width. Its perimeter is 60 m. Find its dimensions and its area. (Very hard.) **[5]**
29. Two cyclists start 120 km apart and ride toward each other, one at 18 km/h and the other at 22 km/h. After how many hours do they meet, and how far has each travelled? (Very hard.) **[6]**

Section D — Communication

30. A store advertises '30% off, then take an extra 20% off the reduced price.' A customer claims this is the same as 50% off. Use a \$100 item to explain why the customer is wrong, and state the single percent discount that is actually equivalent. (Very hard.) **[4]**

— END OF EXAM —